SOCIOL 723: Social Statistics II

Week 5, Potential Outcomes and DAGs

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Today

★ = essential, you must understand this ★ = for understanding, not examined

The Pivot

Weeks 1–4 were about *estimation*: given a target parameter, how do we estimate it and how uncertain are we?

Nothing we did licensed a causal claim. OLS estimates the best linear approximation to a conditional expectation; maximum likelihood estimates the parameters of a probability model. Neither says the word "effect."

Today's question is different

What are we trying to estimate, and what would have to be true about the world for our estimate to recover it?

Two languages answer it, and they are complementary: **potential outcomes** (what would have happened) and **structural equations / directed acyclic graphs** (how the world is wired). Everything in Weeks 6–13 applies one or both.

Potential Outcomes

Two Worlds Per Person ★

For a binary treatment $D_i \in \{0, 1\}$, imagine that each unit i has *two* outcomes written into it before treatment is assigned:

$$Y_i(1) = \text{the outcome if unit } i \text{ were treated,}$$
$$Y_i(0) = \text{the outcome if unit } i \text{ were not treated.}$$

The individual treatment effect is $\tau_i = Y_i(1) - Y_i(0)$.

We observe only one of them:

$$\boxed{Y_i = D_i\, Y_i(1) + (1 - D_i)\, Y_i(0)} \qquad \text{(the } \textit{switching equation}\text{)}$$

Holland's fundamental problem of causal inference

τ_i is never observed for anyone. Causal inference is a missing data problem, and every method in this course is a strategy for filling in the missing column.

The Missing Data View

Unit	D_i	$Y_i(0)$	$Y_i(1)$	τ_i	Y_i (observed)
1	1	?	12	?	12
2	0	7	?	?	7
3	1	?	9	?	9
4	0	5	?	?	5

- Half the table is missing *by construction*, not by bad luck. No larger sample fixes it.
- So we give up on τ_i and target *averages*, which are estimable under assumptions.
- "No causation without manipulation" (Holland 1986): if you cannot even imagine intervening to set D_i, the potential outcomes are not well defined. This is why "the effect of race" or "the effect of gender" needs careful restatement.

Estimands: Say Which Average You Want ★

$$\begin{aligned} \text{ATE} &= E[Y_i(1) - Y_i(0)] && \text{average treatment effect} \\ \text{ATT} &= E[Y_i(1) - Y_i(0) \mid D_i = 1] && \text{on the treated} \\ \text{ATU} &= E[Y_i(1) - Y_i(0) \mid D_i = 0] && \text{on the untreated} \\ \text{CATE}(x) &= E[Y_i(1) - Y_i(0) \mid X_i = x] && \text{conditional on covariates} \end{aligned}$$

- $\text{ATE} = \pi\,\text{ATT} + (1-\pi)\,\text{ATU}$ where $\pi = \Pr(D_i = 1)$.
- These differ whenever effects are heterogeneous *and* correlated with selection, which is the normal case in sociology.
- Policy relevance differs too: "should we expand this programme?" is about the ATU; "was this programme worth running?" is about the ATT.
- Later we will meet estimands nobody chose: the LATE (Week 7), the RD effect at the cutoff (Week 9), and OLS's own variance-weighted average (Week 1).

Estimands, With Numbers

Four units, effects that differ, and treatment correlated with the effect:

Unit	D_i	$Y_i(0)$	$Y_i(1)$	τ_i	Y_i observed
1	1	4	10	6	10
2	1	6	14	8	14
3	0	5	6	1	5
4	0	7	10	3	7

$$\text{ATE} = \tfrac{1}{4}(6+8+1+3) = 4.5, \qquad \text{ATT} = \tfrac{1}{2}(6+8) = 7, \qquad \text{ATU} = \tfrac{1}{2}(1+3) = 2.$$

- The naive comparison is $\frac{1}{2}(10+14) - \frac{1}{2}(5+7) = 12 - 6 = 6$, which equals none of them.

Estimands, With Numbers (cont.)

- Check the decomposition: ATT + selection bias $= 7 + \left(\frac{1}{2}(4+6) - \frac{1}{2}(5+7)\right) = 7 - 1 = 6.$ ✓
- $\text{ATE} = \pi\,\text{ATT} + (1-\pi)\,\text{ATU} = 0.5(7) + 0.5(2) = 4.5.$ ✓

Why the Naive Comparison Fails ★

The quantity we can compute from data is the difference in observed means. Add and subtract $E[Y_i(0) \mid D_i = 1]$:

$$\underbrace{E[Y_i \mid D_i = 1] - E[Y_i \mid D_i = 0]}_{\text{naive comparison}} = E[Y_i(1) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0]$$

$$= \underbrace{E[Y_i(1) - Y_i(0) \mid D_i = 1]}_{\text{ATT}}$$

$$+ \underbrace{E[Y_i(0) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0]}_{\text{selection bias}}.$$

- Selection bias asks: *would the treated and untreated have differed even with no treatment?* Almost always yes.
- Note that it involves only $Y_i(0)$, a quantity about the world as it would have been. No amount of data on Y_i can settle it.

Randomization Solves It ★

If D_i is assigned by a coin flip, then D_i is independent of everything about the unit, including its potential outcomes:

$$(Y_i(0), Y_i(1)) \perp\!\!\!\perp D_i.$$

Then $E[Y_i(0) \mid D_i = 1] = E[Y_i(0) \mid D_i = 0] = E[Y_i(0)]$, the selection bias term is zero, and

$$E[Y_i \mid D_i = 1] - E[Y_i \mid D_i = 0] = \text{ATE} = \text{ATT}.$$

- Randomization does not make the groups identical, it makes them *exchangeable*, so that any difference is chance.
- Note what randomization is doing: it is an assumption made *true by design* rather than argued for.
- Every observational design in this course is an attempt to buy some version of this independence with an argument instead of a coin.

SUTVA: The Assumption Nobody States ★

The framework quietly assumed that writing "$Y_i(1)$" makes sense. That requires the stable unit treatment value assumption:

SUTVA has two parts

1. **No interference**: unit i's outcome does not depend on anyone else's treatment. Otherwise Y_i is a function of the whole vector $(D_1, \ldots, D_n)$ and each unit has 2^n potential outcomes.
2. **No hidden versions of treatment**: "$D_i = 1$" means the same thing for everyone, or $Y_i(1)$ is ambiguous.

- Sociology is full of interference: peer effects, neighbourhood effects, general equilibrium, diffusion. If everyone gets a college degree, the return to a degree changes.
- Hidden versions: "college" covers a community-college certificate and a Princeton BA; "marriage" in 1960 and in 2020.
- State SUTVA explicitly and defend it, or redefine the treatment.

Consistency: the Assumption Hiding in the Notation

The switching equation $Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0)$ contains an assumption so quiet it usually goes unnamed.

Consistency

For a unit that actually received d, the observed outcome *is* that unit's potential outcome under d: $Y_i = Y_i(d)$ when $D_i = d$.

- Sounds like a tautology. It is not: it requires the treatment to be well enough defined that "the outcome under d" is a single quantity.
- It fails when treatment is heterogeneous in ways that matter, "college" that is two years at a community college versus four at a research university, which is the hidden-versions half of SUTVA again.

Consistency: the Assumption Hiding in the Notation (cont.)

- It also fails under *anticipation*: if units respond to the assignment before it is realized, the observed outcome reflects both.
- Consistency is the bridge between the causal notation and the data. Without it, the identification chain has no last step.

Randomization: What It Does and Does Not Buy

- **Does**: makes D_i independent of *everything* about the unit, observed or not, so the selection-bias term is zero in expectation.
- **Does not**: make the two groups identical in your sample. Chance imbalance is real, and it is why we still adjust for pre-treatment covariates in experiments, for *precision*, not for bias.
- **Does not**: rescue you from non-compliance, attrition, or interference. Randomizing the *offer* identifies the effect of the offer (intention-to-treat), not the effect of the treatment, which is the LATE problem of Week 7.
- A balance table with p-values is not a check on randomization. If assignment was truly random, any imbalance is chance by construction, and the p-values are uniform by definition. Report standardized differences instead, and use them to decide what to adjust for.

Identification Under Selection on Observables ★

When randomization is unavailable, we assume it holds *within cells of observed covariates.*

The four conditions

1. **Conditional independence** (unconfoundedness, ignorability): $(Y_i(0), Y_i(1)) \perp\!\!\!\perp D_i \mid X_i$.
2. **Positivity** (overlap): $0 < \Pr(D_i = 1 \mid X_i = x) < 1$ for all x with positive density.
3. **Consistency**: $Y_i = Y_i(D_i)$, so the observed outcome *is* the potential outcome for the treatment actually received.
4. **SUTVA**, as above.

Identification Under Selection on Observables ★ (cont.)

Then the ATE is identified:

$$\text{ATE} = E\Big[\underbrace{E[Y_i \mid D_i = 1, X_i] - E[Y_i \mid D_i = 0, X_i]}_{\text{estimable from data}}\Big].$$

Read the two lines carefully. The left side is causal and unobservable; the right side is a function of the observed joint distribution. Assumptions 1–4 are the bridge, and they are not testable.

What Positivity Is Really Saying ★

$0 < \Pr(D_i = 1 \mid X_i = x) < 1$: within every covariate cell, we must see *both* treated and untreated units.

- If no one with $X = x$ is ever treated, no data in the world tells you what would happen if they were. Extrapolation there is model, not evidence.
- Positivity fails silently when you add many covariates: with 10 binary controls there are 1,024 cells, and most will be empty or one-sided. This is the curse of dimensionality for causal inference, and it is why Weeks 6 and 13 exist.
- Regression does not warn you. It fills empty cells by *extrapolating the functional form*, quietly.
- Diagnose it: look at the distribution of the estimated propensity score in the two groups (Week 6).

Path Analysis to DAGs

Causal Diagrams Are a Sociological Tradition

- The earliest formal treatment of causality in statistics was **Sewall Wright**'s *path analysis* (1918, 1921).
- Sociology adopted it early. **Blau and Duncan**'s *The American Occupational Structure* (1967) made path diagrams of status attainment, father's education and occupation → respondent's education → first job → current occupation, the discipline's central analytic image, and structural equation modelling grew out of it.
- But path analysis had no definition of a causal effect and no rule for deciding which variables to include. The arrows were interpreted, not derived.
- **Judea Pearl** (1995) supplied the missing half: a formal graphical language, plus precise criteria, back-door, front-door, saying *when* an effect is identified from observational data.

Causal Diagrams Are a Sociological Tradition (cont.)

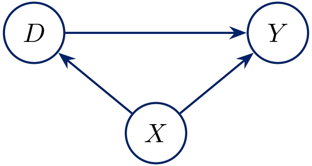
So DAGs are not an import into sociology. They are our own tradition, finally given its identification theory.

What a DAG Is, and What It Leaves Out

Write down how you think each variable is generated. For a treatment D, an outcome Y, and a covariate X:

$$X := \epsilon_X, \qquad D := f_D(X, \epsilon_D), \qquad Y := f_Y(D, X, \epsilon_Y),$$

with the ϵ's mutually independent. Draw an arrow $A \to B$ whenever A appears on the right of B's equation.



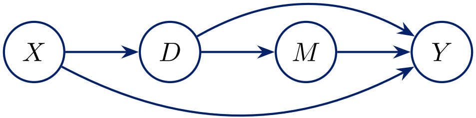
- ":=" means *is generated as*, not *equals*: the equation is a claim about what would happen if we intervened, not a description of a correlation.

What a DAG Is, and What It Leaves Out (cont.)

- **Acyclic**: no variable causes itself through a loop.
- The graph keeps only which arrows exist and drops the functional forms f_D, f_Y. That loss is deliberate, everything below holds for *any* functional form, linear or not.
- The assumptions are the *missing* arrows, not the drawn ones.

The Same Variable Can Play Two Roles

Suppose X also causes D, and D causes M which causes Y.

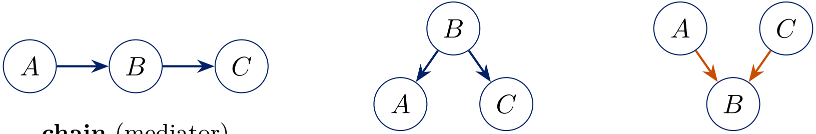


- If the target is the effect of D on Y: X is a **confounder** (control for it) and M is a **mediator** (do not).
- If the target is the *total* effect of X on Y: D is now a **mediator**, and controlling for it destroys the quantity you want.
- Nothing in the data distinguishes these cases. The correct control set depends on which effect you want, so the estimand must be chosen before the specification.

This is why "I controlled for everything available" is not a defensible position. There is no control set that is correct for all questions at once.

DAGs

The Three Building Blocks ★



chain (mediator)

fork (confounder)

collider

Structure	A and C unconditionally	conditioning on B
Chain $A \to B \to C$	associated	blocks the path
Fork $A \leftarrow B \to C$	associated	blocks the path
Collider $A \to B \leftarrow C$	*independent*	opens the path

The collider row is the one that surprises people, and it is the source of most of the damage done by "controlling for everything."

d-Separation ★

Blocking

A path between D and Y is blocked by a set Z if either

1. it contains a chain or fork whose middle node is *in* Z, or
2. it contains a collider that is *not* in Z and none of whose descendants is in Z.

D and Y are d-separated by Z if Z blocks every path between them.

- d-separation is purely graphical; you can check it with a pencil.
- Its content is probabilistic: d-separation by Z implies $D \perp\!\!\!\perp Y \mid Z$ in *every* distribution compatible with the graph.
- Note clause 2's tail: conditioning on a *descendant* of a collider also opens the path, partially. Conditioning on a proxy for a collider is still conditioning on a collider.

The Back-Door Criterion and the *do*-Operator ★

Back-door criterion (Pearl 1995)

A set Z satisfies the back-door criterion relative to the ordered pair (D, Y) if

1. no node in Z is a descendant of D, and
2. Z blocks every path between D and Y containing an arrow into D.

If Z satisfies it, the causal effect is identified by the adjustment formula:

$$P(Y(d)) = P(Y \mid do(D = d)) = \int P(Y \mid D = d, Z = z)\, f(z)\, dz.$$

- $do(D = d)$ means *intervene* and set D to d, as distinct from *observing* $D = d$. Confusing the two is the entire problem.
- In words: average the observed conditional distribution of Y among units with $D = d$ *across strata of* Z, weighted by how common each stratum is in the whole population.

The First Law of Causal Inference ★

Equivalence

The DAG (with its structural equations) and the potential-outcomes framework express the same underlying assumptions in two notations. A graph that satisfies the back-door criterion *delivers* conditional ignorability.

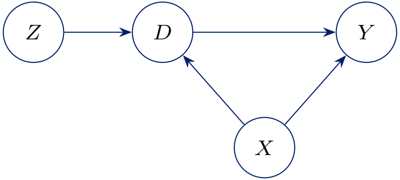
Writing D for the treatment, if Z closes all back doors then

$$Y(d) \perp\!\!\!\perp D \mid Z, \qquad E[Y(d) \mid Z] = E[Y(d) \mid D = d, Z] = E[Y \mid D = d, Z].$$

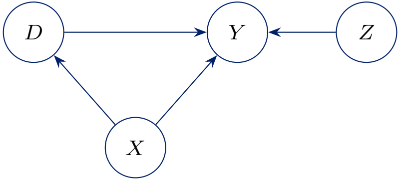
- The last equality is *consistency*: for units that actually received d, the potential outcome $Y(d)$ is the observed Y.
- The chain of three expressions is worth memorizing. The first is causal and unobservable; the last is estimable; the middle step is where the assumption lives.
- Equivalently: conditioning on Z *d-separates* the realized treatment D from the potential outcome $Y(d)$.

Why This Beats “Control for Anything Predictive”

Wooldridge (2005) notes the temptation: “if ‘good’ is taken to mean ‘best fit’, it is tempting to include anything in X_i that helps predict treatment.”



Scenario 1: $Z \to D$ only

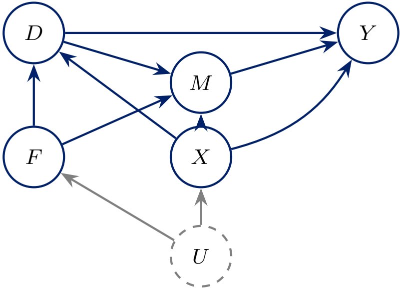


Scenario 2: $Z \to Y$ only

- **Scenario 2** (Z predicts only Y): including Z is a *precision* variable, it soaks up residual variance and shrinks the standard error. Include it.
- **Scenario 1** (Z predicts only D): including Z removes variation in D and inflates the variance, and if any unmeasured confounding remains, it amplifies the bias.

A Worked Example: 401(k) Eligibility and Wealth

Does being *eligible* for a 401(k) plan (D) raise net financial assets (Y)? Eligibility works partly through the employer's matching contribution (M); worker characteristics X, firm characteristics F, and a latent factor U (say, taste for saving) structure the system.

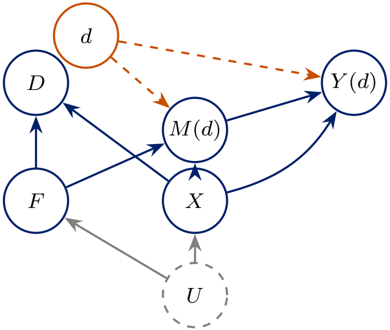


Back-door paths from D to Y: $D \leftarrow X \rightarrow Y$, and $D \leftarrow F \leftarrow U \rightarrow X \rightarrow Y$. Conditioning on $\{F, X\}$ blocks both.

The Same Graph, in Potential-Outcome Notation ★

Conditioning on F and X satisfies the back-door criterion, hence

$$Y(d) \perp\!\!\!\perp D \mid F, X, \qquad E[Y(d) \mid F, X] = E[Y \mid D = d, F, X].$$

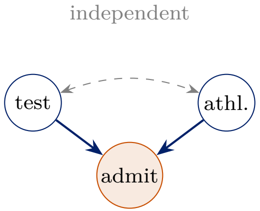


The *realized* treatment D is still a function of F and X, the arrows into D remain. What changes is that $Y(d)$ and $M(d)$ are now fed by the *set* value d (dashed), so no arrow runs from D into them. The picture makes "D is independent of $Y(d)$ given F, X" visible.

Colliders

The Intuition First ★

Admission to a selective college requires either high test scores or athletic talent; the two are independent in the population.



- In the population, test scores tell you nothing about athletic talent.

The Intuition First ★ (cont.)

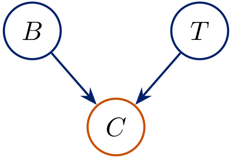
- Among admitted students: someone with mediocre scores must have got in some other way, so they are probably a strong athlete. A *negative* association appears out of nowhere.
- Berkson's paradox, the same mechanism behind "selection bias" and the birth-weight paradox.

Now the Algebra ★

Consider the structural model

$$T := \epsilon_T, \qquad B := \epsilon_B, \qquad C := T + B + \epsilon_C,$$

with $\epsilon_T, \epsilon_B, \epsilon_C$ independent $\mathcal{N}(0,1)$.



- B and T are independent by construction: $E[T] = 0$ and $E[T \mid B = b] = 0$ for every b.
- There is no arrow between them, and no common cause.
- Question: what is $E[T \mid B, C]$?

Conditioning on the Collider ★

Because $C = T + B + \epsilon_C$, we can write $T = (C - B) - \epsilon_C$. So $E[T \mid B, C]$ is the linear projection of T on $C - B$:

$$E[T \mid B, C] = \frac{\operatorname{Cov}\left(T,\ C - B\right)}{\operatorname{Var}\left(C - B\right)}(C - B) = \frac{\operatorname{Cov}\left(T,\ T + \epsilon_C\right)}{\operatorname{Var}\left(T + \epsilon_C\right)}(C - B)$$

$$= \frac{1}{1+1}(C - B) \ = \ \tfrac{1}{2}(C - B).$$

Conditioning on the Collider ★

Because $C = T + B + \epsilon_C$, we can write $T = (C - B) - \epsilon_C$. So $E[T \mid B, C]$ is the linear projection of T on $C - B$:

$$E[T \mid B, C] = \frac{\text{Cov}\left(T,\ C - B\right)}{\text{Var}\left(C - B\right)}\,(C - B) = \frac{\text{Cov}\left(T,\ T + \epsilon_C\right)}{\text{Var}\left(T + \epsilon_C\right)}\,(C - B)$$

$$= \frac{1}{1 + 1}\,(C - B) \;=\; \tfrac{1}{2}(C - B).$$

Read the coefficient on B straight off that conditional expectation:

$$E[T \mid B, C] = \tfrac{1}{2}C \;-\; \tfrac{1}{2}B \qquad \Longrightarrow \qquad \frac{\partial\, E[T \mid B, C]}{\partial B} = -\tfrac{1}{2}.$$

Conditioning on the Collider: What It Means

Controlling for C, the apparent effect of B on T is $-1/2$: entirely spurious, since nothing in the model connects B to T, and indeed $E[T \mid B] = 0$ for every B.

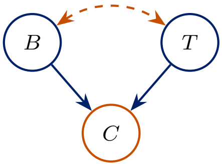
This is collider bias, the same object as Heckman's sample-selection bias.

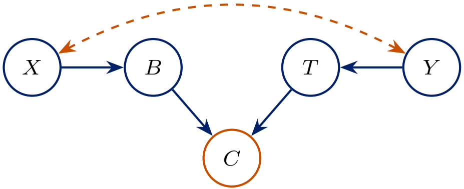
Read the *partial* coefficient, not the average

Averaging $\frac{1}{2}(C - b)$ over C given $B = b$ gives zero, because $E[C \mid B = b] = b$. The bias lives in the comparison at *fixed* C, which is exactly what "controlling for C" does.

The Damage Spreads to Ancestors

Conditioning on a collider opens associations among its parents and all of their ancestors.





- On the right, X and Y are four steps apart and have no common cause. Conditioning on C makes them dependent anyway.
- Practical consequence: a control variable can induce bias between variables it is not adjacent to. You cannot check for this by inspecting correlations, only by tracing paths.
- And conditioning on a *descendant* of C does a milder version of the same thing.

The Birth-Weight Paradox

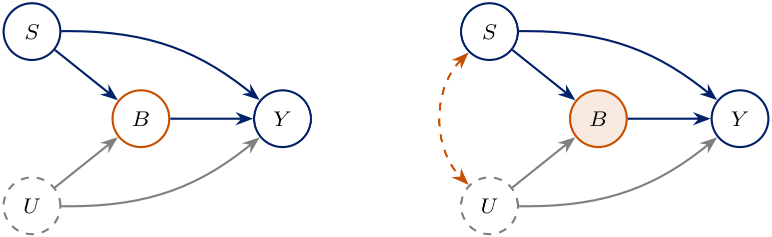
A famous puzzle from a 1991 US study.

- Infants born to smokers have a higher risk of low birth weight (LBW).
- Infants born to smokers have higher infant mortality.
- But among LBW infants, mortality is *lower* for infants of smokers than for infants of non-smokers.

The naive reading is that smoking is protective conditional on LBW, which would be a remarkable public-health finding.

The plausible reading is that **low birth weight is a collider**. Both smoking and unobserved competing risks (congenital abnormalities, severe maternal illness) cause LBW, and those competing risks are far more lethal than smoking. Among LBW babies, "not a smoker" becomes evidence for "something worse."

The Birth-Weight Paradox, Structurally ★



$$Y := S + B + \kappa U + \epsilon_Y, \qquad B := S + U + \epsilon_B, \qquad S := \epsilon_S, \qquad U := \epsilon_U,$$

with independent $\mathcal{N}(0,1)$ shocks. Smoking has a *harmful* total effect: direct 1, plus indirect 1×1 through birth weight, total $\mathbf{+2}$.

Where the Sign Flips ★

Projecting Y on S alone recovers the correct total effect of $+2$.

Projecting Y on S *and* B, "controlling for birth weight", gives the conditional expectation

$$E[Y \mid S, B] \;=\; \big(1-\kappa/2\big)\,S \;+\; \big(1+\kappa/2\big)\,B.$$

- The coefficient on S is no longer the direct effect of 1. It has acquired a $-\kappa/2$ term, which comes entirely from the induced negative association between S and U within strata of B.
- If the competing risks are much more lethal than smoking, $\kappa \gg 1$, the coefficient on smoking becomes large and negative. Smoking appears strongly protective.
- Nothing is wrong with the estimator. The regression faithfully reports a conditional expectation. The error is in reading that conditional expectation as a causal effect.

Selection Bias Is Collider Bias ★

The most common invisible collider in sociology is *the sample itself.*

- **Wage regressions on employed workers**: employment is caused by both schooling and the unobserved determinants of wages. Restricting to the employed conditions on a collider.
- **Studies of college students, surviving firms, intact marriages, surviving states**: every one is a conditioning operation.
- **Panel attrition**: staying in the sample is an outcome.
- **"Among those who report X"**: reporting is behaviour.

The practical rule

Draw the selection mechanism into your DAG as a node you have conditioned on, with a box around it. Then read off whether it opens anything. If your sample was constructed by any process downstream of treatment or outcome, assume it did.

Controls

The Taxonomy ★

Variable	Role	Control for it?
Common cause of D and Y	confounder	**yes**; this is the point
Pre-treatment predictor of Y only	precision variable	**yes**, reduces variance
Pre-treatment predictor of D only	instrument-like	*no*, inflates variance, amplifies bias
Consequence of D on the path to Y	mediator	no, removes part of the effect
Common effect of D and Y	collider	no, creates spurious association
Consequence of Y	outcome proxy	no
Sample selection on a collider	implicit conditioning	no, and often invisible

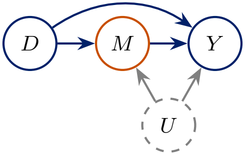
- Full taxonomy in Cinelli, Forney, and Pearl (2024).
- Note the third row again: conditioning on a strong predictor of D with no direct effect on Y *amplifies* whatever unmeasured confounding remains. "More controls" is not conservative.

Post-Treatment Variables, and How They Sneak In

- Explicitly adjusting for a post-treatment variable is almost always a mistake, and it is a mistake even when the coefficient "looks better."
- The greater danger is implicit conditioning, through sample construction and variable definition rather than through the regression formula:
  - estimating the effect of education on wages using only *employed* individuals, employment is post-treatment;
  - studying occupational outcomes among people who *have* an occupation;
  - defining the outcome as a ratio whose denominator responds to treatment;
  - restricting to respondents who completed a follow-up wave.
- None of these appears in the model equation. All of them are conditioning.

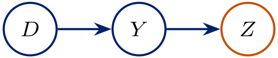
Two Sharper Cases

Mediators and the controlled direct effect



M is a bad control *even if you want the controlled direct effect*: conditioning on M opens the collider path $D \to M \leftarrow U \to Y$. Direct effects need *stronger* assumptions than total effects.

Consequences of the outcome



Z happens *after* Y and is caused by it. Controlling for it partials out variation in the outcome itself and biases the estimate. Which direction depends on the rest of the graph, so do not assume it is conservative. Common as "controlling for" a downstream measure of the same construct.

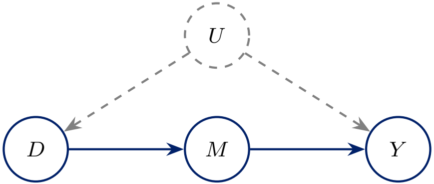
Two Sharper Cases (cont.)

No statistical test distinguishes any of these from a confounder. The distinction is causal, and it must come from the DAG.

Front Door

Identification Without Measuring the Confounder

Sometimes no back door can be closed, but the *mechanism* can be measured.



The three conditions

1. **Complete mediation**: every directed path from D to Y passes through M.
2. **No unblocked back door from D to M.**
3. **All back doors from M to Y are blocked by D.**

U is unobserved and confounds D and Y, yet the effect is identified. That is a genuinely surprising result, and worth understanding rather than memorizing.

Why It Works: Two Identified Legs

The trick is to split $D \to Y$ into two effects, each of which *is* identified.

- **Leg 1,** $D \to M$: by condition 2 there is no back door, so $P(M \mid do(D = d)) = P(M \mid D = d)$. We can read the effect of D on M straight off the data.
- **Leg 2,** $M \to Y$: the only back door from M to Y is $M \leftarrow D \leftarrow U \to Y$, and conditioning on D blocks it. So the effect of M on Y is identified by adjusting for D.
- **Compose**: the effect of D on Y is the effect of D on M, pushed through the effect of M on Y.

Neither leg requires observing U. The confounder sits outside both sub-problems, and that is the whole trick.

The Front-Door Formula ★

$$
\begin{aligned}
P(Y(d)) &= P(Y \mid do(D=d)) \\
&= \sum_m P(Y \mid do(D=d), M=m)\, P(M=m \mid do(D=d)) \\
&= \sum_m \underbrace{P(Y \mid do(M=m))}_{\text{leg 2}}\ \underbrace{P(M=m \mid D=d)}_{\text{leg 1}} \\
&= \sum_m P(M=m \mid D=d) \left[\sum_{d'} P(Y \mid M=m, D=d')\, P(D=d')\right].
\end{aligned}
$$

- Line 3 uses complete mediation (given M, D no longer matters for Y) and condition 2.

The Front-Door Formula ★ (cont.)

- The bracket in line 4 is the back-door adjustment for leg 2, averaging over the *marginal* distribution of D, note d', not d.
- Everything on the right is estimable. U never appears.

A Sociological Application: the Age–Period–Cohort Problem

To see what front-door reasoning actually buys, take a problem sociologists know well, and one where the regression coefficient cannot even be *defined* uniquely: separating age, period, and cohort.

The standard APC regression is

$$Y_{it} = \alpha + \beta_A A_i + \beta_P P_t + \beta_C C_i + \epsilon_{it},$$

with A = age, P = calendar period, C = birth cohort.

The identification problem

$C = P - A$ exactly. The three regressors are *perfectly collinear*, so $\beta_A, \beta_P, \beta_C$ are not separately identified, for any candidate solution there is a one-parameter family of alternatives fitting equally well.

A Sociological Application: the Age–Period–Cohort Problem (cont.)

Decades of proposed fixes, constrained estimators, the intrinsic estimator, arbitrary equality restrictions, amount to choosing a point on that family by assumption. None of them is identification; they are conventions.

Mechanism-Based Identification for APC

Suppose we want the effect of *historical period* on attitudes. Rather than fight the collinearity, look for the mechanism.

- Period (D) affects exposure to new information, institutions, media environments, or policies (M).
- Those exposures shape individual attitudes (Y).
- If we can measure *all* the channels M through which period operates, the front-door formula identifies the period effect:

$$P(Y \mid do(D)) = \sum_m P(M = m \mid D) \sum_{d'} P(Y \mid M = m, D = d')P(D = d').$$

The collinearity is not the binding constraint once we stop asking for a coefficient in a linear model and start asking for a causal quantity with a stated mechanism.

Critiques of Front-Door Identification

Be sceptical of front-door claims, including your own.

- **Complete mediation is unrealistic.** Every effect of D on Y must pass through the *observed* M. A single unmediated path breaks identification, and in the APC case period plausibly acts through many unmeasured channels at once.
- **No hidden confounding of the legs.** We still need $D \to M$ unconfounded, and $M \to Y$ identified by adjusting for D alone. If something unobserved drives both the mechanism and the outcome, the second leg fails.
- **Measurement.** M must be measured well, and note *where* the error bites. In leg 1, M is the outcome, so classical error adds noise without attenuating $P(M \mid D)$. In leg 2, M is the variable being intervened on and conditioned on, so error there biases that leg and therefore the composed estimate.

Critiques of Front-Door Identification (cont.)

- **Sensitivity.** Because the assumptions are strong, report how the estimate moves as you allow a small unmediated path.

The front door is a real result and a rarely satisfied one. Its main value is clarifying *what you would need to know* to make a mechanism argument work.

Potential Outcomes and DAGs Are Complements

Potential outcomes

- Precise about the *estimand*: ATE, ATT, LATE.
- Natural home for heterogeneity, effect distributions, and design-based inference.
- Clumsy for reasoning about long chains of variables.

DAGs

- Precise about *identification*: which paths are open, what to condition on.
- Makes collider and mediator mistakes visible immediately.
- Silent about magnitude, heterogeneity, and which average you get.

Use both

Draw the DAG to work out *what to condition on.* Write the potential outcomes to state *what you are estimating.* The First Law says they are the same theory; in practice each makes a different mistake easy to catch.

Estimands

The Standard Workflow Skips a Step ★

Lundberg, Johnson, and Stewart (2021) argue that most quantitative papers jump straight from theory to a regression, leaving the target quantity implicit. Their proposed sequence:

1. **Theoretical estimand.** A *unit-specific quantity* and a *target population*. "The average difference in earnings at age 40 between completing college and not, among US-born adults who completed high school." Stated before any data.
2. **Empirical estimand.** What identification assumptions turn the theoretical estimand into a function of observable distributions? This is the back-door argument.
3. **Estimation strategy.** Only now: regression, matching, weighting, IV, machine learning.

Most methodological disputes are really disagreements at step 1 or 2 being conducted as if they were about step 3.

Why This Discipline Pays

- It forces you to notice when the estimand is chosen *by the method* rather than by you: OLS's variance-weighted average (Week 1), the LATE for compliers (Week 7), the effect at the cutoff (Week 9).
- It makes "the coefficient on X_3 in Model 4", a quantity nobody can state in words, an unacceptable answer.
- It separates two very different objections to a paper: "your estimand is uninteresting" and "your estimate does not recover your estimand."
- It clarifies external validity: the target population is part of the estimand, so extrapolation is a stated claim rather than an implicit one.

Why This Discipline Pays (cont.)

A habit worth forming

Write the theoretical estimand as one sentence, in words, before you write any code. You will be asked to do exactly this on the final exam, and in every paper you referee afterwards.

Where This Leaves Us

- Causal inference is a missing data problem. We target averages because individual effects are unknowable.
- The naive comparison equals the ATT plus selection bias. Every design in this course is an argument *about* that term: either that adjustment drives it to zero (Week 6), or that some other comparison sidesteps it entirely, at the price of a narrower estimand (IV's LATE, Weeks 7–11).
- Conditional ignorability, positivity, consistency, and SUTVA identify the ATE from observable quantities. None of them is testable.
- DAGs turn those assumptions into a picture you can check for colliders and mediators.

Where This Leaves Us (cont.)

Coming next

Week 6: if we believe conditional ignorability, how should we actually adjust? Matching, propensity scores, weighting, and doubly robust estimation, and why regression is only one option among several.

Weeks 7–11: what to do when we do *not* believe it.

References

- Angrist, Joshua D. and Jörn-Steffen Pischke. 2009. *Mostly Harmless Econometrics*. Princeton University Press. [Ch. 2]
- Cinelli, Carlos, Andrew Forney, and Judea Pearl. 2024. "A Crash Course in Good and Bad Controls." *Sociological Methods & Research* 53(3): 1071–1104.
- Greenland, Sander and Judea Pearl. 2017. "Causal Diagrams." *Wiley StatsRef: Statistics Reference Online*, 1–10.
- Holland, Paul W. 1986. "Statistics and Causal Inference." *Journal of the American Statistical Association* 81(396): 945–960.
- Lundberg, Ian, Rebecca Johnson, and Brandon M. Stewart. 2021. "What Is Your Estimand? Defining the Target Quantity Connects Statistical Evidence to Theory." *American Sociological Review* 86(3): 532–565.
- Morgan, Stephen L. and Christopher Winship. 2015. *Counterfactuals and Causal Inference*, 2nd ed. Cambridge University Press. [Ch. 1–3]
- Pearl, Judea. 1995. "Causal Diagrams for Empirical Research." *Biometrika* 82(4): 669–688.

References (cont.)

- Rubin, Donald B. 1974. "Estimating Causal Effects of Treatments in Randomized and Nonrandomized Studies." *Journal of Educational Psychology* 66(5): 688–701.